

Zihan Wang

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Engine Developer / Graphics Rendering Engineer

SUMMARY

Master's graduate in Computer Science with a B.S. in Digital Media Technology and hands-on experience in VR development at NetEase, game development at Alibaba, and Unreal Engine VR development at RIT MAGIC Spell Studios. Core strengths include C++ engine development and real-time graphics, with experience in Vulkan/OpenGL rendering pipelines, Render Graph architecture, resource management, GPU debugging, and performance profiling.

TECHNICAL SKILLS

Programming Languages: C++, C#, Python, Java

Graphics APIs: Vulkan, OpenGL

Shader Languages: GLSL, HLSL

Game Engines: Unreal Engine 5, Unity

Rendering: Render Graph, PBR, shadow mapping, IBL, HDR, tone mapping

Profiling and Debugging: RenderDoc, Vulkan Validation Layers, GPU profiling

XR and Computer Vision: Meta SDK, OpenXR, ARCore, ARKit, PyTorch, OpenCV, 3D Gaussian Splatting

Development Tools: Git, Docker, Blender, Maya, Jira

EXPERIENCE

RIT MAGIC Spell Studios - VR Developer (Unreal Engine)

Aug 2025 - May 2026

Unreal Engine, C++, Blueprint

- Developed Unreal Engine VR prototypes and immersive research applications in C++ and Blueprint, contributing interaction systems, character and environment modules, and cross-system integration.
- Diagnosed and mitigated tracking jitter, state synchronization, feedback continuity, and runtime stability issues to support iterative user testing and live project demonstrations.

NetEase, Guangzhou - VR Development Intern

Mar 2023 - Aug 2023

VR Client Systems, Interaction, and Performance Optimization

Unity, Meta Quest, PICO

- Prototyped core VR gameplay for Meta Quest and PICO in Unity, implementing interaction systems, input feedback, and scene logic.
- Profiled rendering, scripting, and interaction paths with Unity Profiler and runtime diagnostics, mitigating frame drops and improving runtime stability.

Alibaba, Hangzhou - Unity Game Development Intern

Jun 2022 - Aug 2022

Game Systems and Resource Management

Unity

- Implemented Unity gameplay and core-system features spanning inventory, configuration loading, and resource management.
- Refactored data organization, lookup logic, and resource references to improve memory efficiency, access performance, and maintainability.

SELECTED PROJECTS

Vulkan Engine Prototype / Real-Time Renderer

Sep 2025 - May 2026

C++20, Vulkan 1.3, GLSL, Render Graph

- Engineered a Vulkan 1.3 renderer in C++20 with swapchain management, command buffers, descriptors, pipelines, dynamic rendering, and Synchronization2.
- Implemented an end-to-end real-time rendering path with PBR, normal mapping, PCF shadows, HDR, bloom, tone mapping, IBL prefiltering, and BRDF LUT generation.
- Architected a Render Graph around explicit per-pass resource declarations to manage image layout transitions, transient render targets, resource lifetimes, and barrier generation. Added GPU timing instrumentation and render-target inspection tools.

CGRenderEngine - OpenGL Rendering Engine

Jan 2024 - Aug 2024

C++, OpenGL, GLSL

- Architected a modular OpenGL rendering engine from the ground up, covering scene organization, shader and material systems, texture resources, lighting, shadows, HDR, and post-processing.
- Implemented PBR, shadow mapping, framebuffer post-processing, and tone mapping, then extended the engine with ray tracing and supersampling experiments to compare real-time rasterization against offline rendering.

Kulla-Conty Multiple-Scattering BRDF in UE5

Jan 2025 - May 2025

C++, HLSL, Unreal Engine 5

- Implemented Kulla-Conty multiple-scattering energy compensation for GGX BRDFs to address energy loss at high roughness.
- Developed a Monte Carlo GGX integrator to generate compensation LUTs and integrated the generated LUT into UE5 materials through custom HLSL nodes.

EDUCATION

Rochester Institute of Technology (RIT)

Aug 2023 - May 2026

Master of Science in Computer Science. Relevant Coursework: Computer Graphics, Algorithms, Machine Learning, Computer Vision, Advanced Object-Oriented Programming

Xiamen University

Sep 2019 - Jun 2023

Bachelor of Science in Digital Media Technology. Relevant Coursework: C/C++, Data Structures, Operating Systems, Game Engine Design, Game Development